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ORIGINAL CONTRIBUTION

# Increased Risk of Ischemic Events in Patients With Stroke Treated With Clopidogrel and a P-CAB or PPI

Jung-Hwa Hong*; Seong-Hun Park*; Kwang-Yeol Park, MD, PhD; Eun-Young Kim, PhD

**BACKGROUND:** Potassium-competitive acid blockers (P-CABs) are newly introduced gastrointestinal protectants, but their interaction with clopidogrel remains uncharacterized. Proton pump inhibitors (PPIs) are often coprescribed with clopidogrel, despite potential pharmacodynamic interactions. This study evaluated the effectiveness and safety of P-CAB and PPI coadministered with clopidogrel in patients with ischemic stroke.

**METHODS:** This population-based, longitudinal, retrospective cohort study used the Korean National Health Insurance Service database, covering 51.4 million individuals (≈97% of the Korean population) from 2016 to 2022. Patients with incident ischemic stroke were identified using predefined criteria. Propensity score matching compared the 6-month incidence of major adverse cardiovascular events (including myocardial infarction, stroke recurrence, and all-cause mortality) and gastrointestinal bleeding among 3 cohorts: clopidogrel alone, clopidogrel+P-CAB, and clopidogrel+PPI.

**RESULTS:** Between 2017 and 2021, 40 752 of 65 180 patients with incident stroke (62.5%) received clopidogrel. A total of 6680 patients (16.4%) formed the clopidogrel-alone cohort, 578 (1.4%) the clopidogrel+P-CAB cohort, and 8806 (21.6%) the clopidogrel+PPI cohort. During the follow-up period, stroke recurred in 268 patients (4.0%) receiving clopidogrel alone, 26 patients (4.5%) receiving clopidogrel+P-CAB, and 431 patients (4.9%) receiving clopidogrel+PPI. After propensity score matching, the clopidogrel+P-CAB cohort showed a higher risk of major adverse cardiovascular events (hazard ratio [HR], 2.40 [95% CI, 1.18–4.85]) and stroke recurrence (HR, 2.64 [95% CI, 1.27–5.47]) than the clopidogrel-alone cohort. The clopidogrel+PPI cohort also had an increased risk of major adverse cardiovascular events (HR, 1.38 [95% CI, 1.17–1.63]) and stroke recurrence (HR, 1.41 [95% CI, 1.20–1.67]) relative to clopidogrel alone. Among PPIs, esomeprazole was associated with a higher risk of major adverse cardiovascular events (HR, 1.46 [95% CI, 1.10–1.92]) and stroke recurrence (HR, 1.50 [95% CI, 1.13–2.00]). Gastrointestinal bleeding did not differ significantly among the clopidogrel-alone, clopidogrel+P-CAB, and clopidogrel+PPI cohorts.

**CONCLUSIONS:** Concomitant use of P-CABs or PPIs with clopidogrel was associated with an increased risk of ischemic events in patients with stroke. P-CABs or PPIs should, therefore, be prescribed cautiously for patients with stroke treated with clopidogrel.

**GRAPHIC ABSTRACT:** A graphic abstract is available for this article.

**Key Words:** cause of death ■ clopidogrel ■ outpatients ■ proton pump inhibitors ■ stroke

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Stroke is one of the leading causes of death and long-term disability.[1] In patients with ischemic stroke, clopidogrel is commonly prescribed for secondary prevention, and it is often coprescribed with proton pump inhibitors (PPIs), especially when used together with aspirin or in those with a high risk of gastrointestinal bleeding.[2] PPIs, however, may inhibit the formation of the active clopidogrel metabolite via the CYP2C19 enzyme system. In 2011, the US Food and Drug Administration (FDA)

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Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.125.053616.
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## Nonstandard Abbreviations and Acronyms

| | |
|---|---|
| **CYP** | cytochrome P450 |
| **FDA** | Food and Drug Administration |
| **ICD-10** | International Classification of Diseases, Tenth Revision |
| **HR** | hazard ratio |
| **KNHIS** | Korean National Health Insurance Service |
| **MACE** | major adverse cardiovascular events |
| **MI** | myocardial infarction |
| **P-CAB** | potassium-competitive acid blocker |
| **PPI** | proton pump inhibitor |
| **PSM** | propensity score matching |

advised against the coadministration of clopidogrel with omeprazole.[3]

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Recently developed potassium-competitive acid blockers (P-CABs) provide gastrointestinal protection comparable to PPIs, and their use as PPI alternatives is increasing in clinical practice.[4,5] The nationwide South Korean health insurance claims database has captured these prescribing patterns, showing that P-CABs and PPIs are commonly coadministered with clopidogrel in real-world stroke care. Vonoprazan, a representative P-CAB, received FDA approval in November 2023. Tegoprazan, approved by the Korean FDA in 2018, is currently undergoing US clinical trials under the name BLI5100 (Unique identifiers: NCT05890001, NCT05587322, and NCT05587309), and fexuprazan was approved in South Korea in 2021.

Despite the increasing availability of these new P-CABs, robust clinical outcome data for clopidogrel combined with P-CABs remain lacking. For instance, in a randomized, open-label, 2-period crossover trial, vonoprazan was shown to achieve faster and more prolonged gastric pH control than PPIs.[6] Furthermore, another study examining the acid-inhibitory effects of vonoprazan versus esomeprazole stratified by CYP2C19 genotype also showed that vonoprazan provided significantly more potent acid suppression[7]; however, a separate, small study reported that vonoprazan may lower clopidogrel's active metabolite levels and attenuate antiplatelet effects,[8,9] but comprehensive evidence is scarce.[10] In addition, evidence regarding the safety and efficacy of clopidogrel with PPIs in patients with stroke remains inconsistent, often varying depending on the specific PPI used.[3,11–14] To date, no large-scale real-world study has evaluated the effectiveness and safety of clopidogrel when combined with either P-CABs or PPIs in this population.

Therefore, this study aimed to assess the real-world clinical outcomes of clopidogrel coadministered with P-CABs or PPIs in patients after stroke, with particular focus on individual PPI agents.

# METHODS

## Data Source

This retrospective cohort study was performed using the Korean National Health Insurance Service (KNHIS) customized database from 2016 to 2022, the complete period available. The KNHIS database includes comprehensive health information on about 51.4 million individuals, representing ≈97% of the Korean population. It contains demographic characteristics, diagnoses, direct medical expenses, prescription records, procedures, and inpatient and outpatient claims. The customized data set provides complete, date-stamped claims for hospitals, clinics, and pharmacies, including eligibility data, medical statements, disease and treatment codes, and prescription details for the investigator-specified period. Claims are coded using the *Korean Standard Classification of Diseases, Seventh Revision*, an adaptation of the *International Classification of Diseases, Tenth Revision* (*ICD-10*). Drug codes follow the Korean Standard Drug Classification system.

## Identification of Incident Stroke

Patients with incident stroke were selected according to the following criteria: (1) adults aged ≥20 years who received a first-ever diagnosis of ischemic stroke (cerebral infarction, *ICD-10* code I63) between January 2017 and December 2021, excluding those with any ischemic stroke diagnosis recorded ≥1 year before the index event; and (2) among these patients, only those admitted through the emergency department or an outpatient clinic who underwent brain imaging (computed tomography or magnetic resonance imaging) during that admissions were included. The follow-up period was 180 days. Figure S1 illustrates the study design.

The following exclusion criteria were applied to the incident stroke cohort: (1) a history of atrial flutter or atrial fibrillation, myocardial infarction (MI) within the previous year, or MI treated with percutaneous coronary intervention or coronary artery bypass grafting within 30 days after the incident stroke, to avoid antiplatelet-related confounding; (2) receipt of heparin, low-molecular-weight heparin, or any oral anticoagulant, owing to the increased risk of bleeding; (3) death within 30 days of the index admission, to minimize immortal-time bias; and (4) receipt of antiplatelet or antiulcer therapy for fewer than 30 days after the index stroke event, to ensure adequate medication exposure and account for cumulative dose.

## Exposure to Medications (Clopidogrel, PPIs, or P-CABs)

The use of clopidogrel, P-CABs, and PPIs was confirmed through detailed prescription records from the National Health Insurance Service. Medication adherence was primarily assessed using the medication possession ratio, defined as the total days of medication supplied by prescriptions divided by the period from the stroke index date to major adverse cardiovascular events (MACE) or the end of follow-up (Tables S1 and S2).

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P-CAB exposure included tegoprazan, fexuprazan, vonoprazan, and revaprazan, whereas PPI exposure comprised esomeprazole, (dex)lansoprazole, omeprazole, pantoprazole, ilaprazole, and rabeprazole. Patients with incident stroke treated with clopidogrel were, therefore, classified into the following cohorts: clopidogrel alone, clopidogrel+P-CAB, and clopidogrel+PPI. Table S3 details the drug-classification algorithm.

## Outcomes and Covariates

All outcomes were captured using a validated claims-based algorithm (Table S4) and were assessed during a 180-day follow-up, consistent with previous risk-window definitions.[10,15]

The primary outcome was the occurrence of MACE, including MI, stroke recurrence, and all-cause mortality. MI was identified by *ICD-10* codes I21.x and I22.x. Stroke recurrence was defined as readmission for ischemic stroke with *ICD-10* code I63 through the emergency department or an outpatient clinic during the follow-up period, accompanied by brain imaging (computed tomography or magnetic resonance imaging) at readmission.[16] All-cause mortality was identified by *ICD-10* codes I46, R96.x, R98, and R99, and KNHIS-Death data. Gastrointestinal bleeding was identified by *ICD-10* codes K25.x, K26.x, K27.x, K28.x, K29.x, and K92.x.[16]

Demographic and clinical characteristics, including comorbidities and concomitant medications, were assessed within the 12 months preceding the index stroke. The Charlson Comorbidity Index was calculated for each patient, encompassing 17 comorbidity categories defined by Quan et al Individual comorbidities were scored 0, 1, or ≥2 based on mortality risk, and the sum yielded the Charlson Comorbidity Index.

## Statistical Analysis

The distributions of measurement variables and baseline characteristics were summarized descriptively. For continuous variables, mean±SD values are presented; for categorical variables, counts and percentages are reported. All statistical tests were 2-sided with a 5% significance level. Covariates entered into the propensity score model included demographic factors (sex and age); comorbidities associated with vascular risk (hypertension, diabetes, dyslipidemia, angina, and hemorrhagic stroke); factors related to gastrointestinal bleeding (gastrointestinal bleeding, gastroesophageal reflux disease, peptic ulcer, gastritis and duodenitis, and aspirin use); and comorbidities associated with overall prognosis (Charlson Comorbidity Index, chronic obstructive pulmonary disease, heart failure, and chronic renal failure).[17–19] We utilized propensity score matching (PSM) to create balanced study groups and effectively account for potential confounding variables. The estimation of the propensity score was performed using a multivariable logistic regression model, which included all relevant baseline covariates to calculate the probability of treatment exposure for each patient. For the matching procedure, we used the one-to-one greedy nearest-neighbor matching without replacement algorithm, strictly maintaining a 1:1 matching ratio and setting a caliper of 0.001 to ensure close matches.[20] The overall quality and balance of the matched sample were rigorously assessed post-matching. We calculated the standardized mean difference for all individual covariates; a value of <0.1 was required for each covariate to confirm that an adequate balance was achieved.[21] In addition, prespecified subgroup analyses were conducted to assess the effectiveness and safety of clopidogrel according to concomitant use of specific types of PPIs (omeprazole, …, ilaprazole/rabeprazole) versus no PPI use. Finally, we performed disaggregated analyses stratified by sex. Using the matched data set, hazard ratios (HRs) and 95% CIs were computed for all outcomes. Analyses were performed with SAS Enterprise Guide (SAS Institute, Cary, NC).

## Ethical Considerations

This study was approved by the institutional review board of Chung-Ang University (institutional review board number 1041078-20230224-HR-038) and the KNHIS Medical Information Disclosure Committee (REQ202300925-002). All procedures conformed to relevant guidelines and regulations. As all data were anonymized and collected retrospectively, the institutional review board granted a waiver of informed consent. This study adheres to the Strengthening the Reporting of Observational Studies in Epidemiology Statement.

# RESULTS

## Baseline Characteristics

The Figure depicts the cohort-selection process. The KNHIS-customized database facilitated the identification of 192 923 patients with incident stroke aged ≥20 years who were treated between January 2017 and December 2021. Among them, 96 665 were hospitalized after presenting to the emergency department or an outpatient clinic and underwent computed tomography or magnetic resonance imaging. After applying exclusion criteria, 65 180 patients with incident stroke remained. Of these, 58 690 received antiplatelet or anticoagulant therapy, and 40 752 were prescribed clopidogrel. A total of 6680 patients (16.39%) comprised the clopidogrel-alone cohort, 578 (1.42%) the clopidogrel+P-CAB cohort, and 8806 (21.61%) the clopidogrel+PPI cohort. Stroke recurred in 268 patients (4.01%) in the clopidogrel-alone cohort, 26 (4.50%) in the clopidogrel+P-CAB cohort, and 431 (4.89%) in the clopidogrel+PPI cohort.

Table 1 summarizes demographic and clinical characteristics before matching. The median age of the patients was 66.44 years, and 58.01% were male. The mean Charlson Comorbidity Index score was 1.40 (±0.88). After 1:1 PSM, the clopidogrel+P-CAB cohort included 536 patients, and the clopidogrel+PPI cohort included 5847 patients. Table 2 confirms that covariates were balanced after 1:1 PSM for variables potentially influencing MACE and gastrointestinal bleeding.

## Clopidogrel+P-CAB Cohort

Table 3 shows that in the PSM analysis, MACE occurred significantly more frequently in the clopidogrel+P-CAB cohort than in the clopidogrel-alone cohort (HR, 2.40 [95% CI, 1.18–4.85]). In addition, stroke recurrence was

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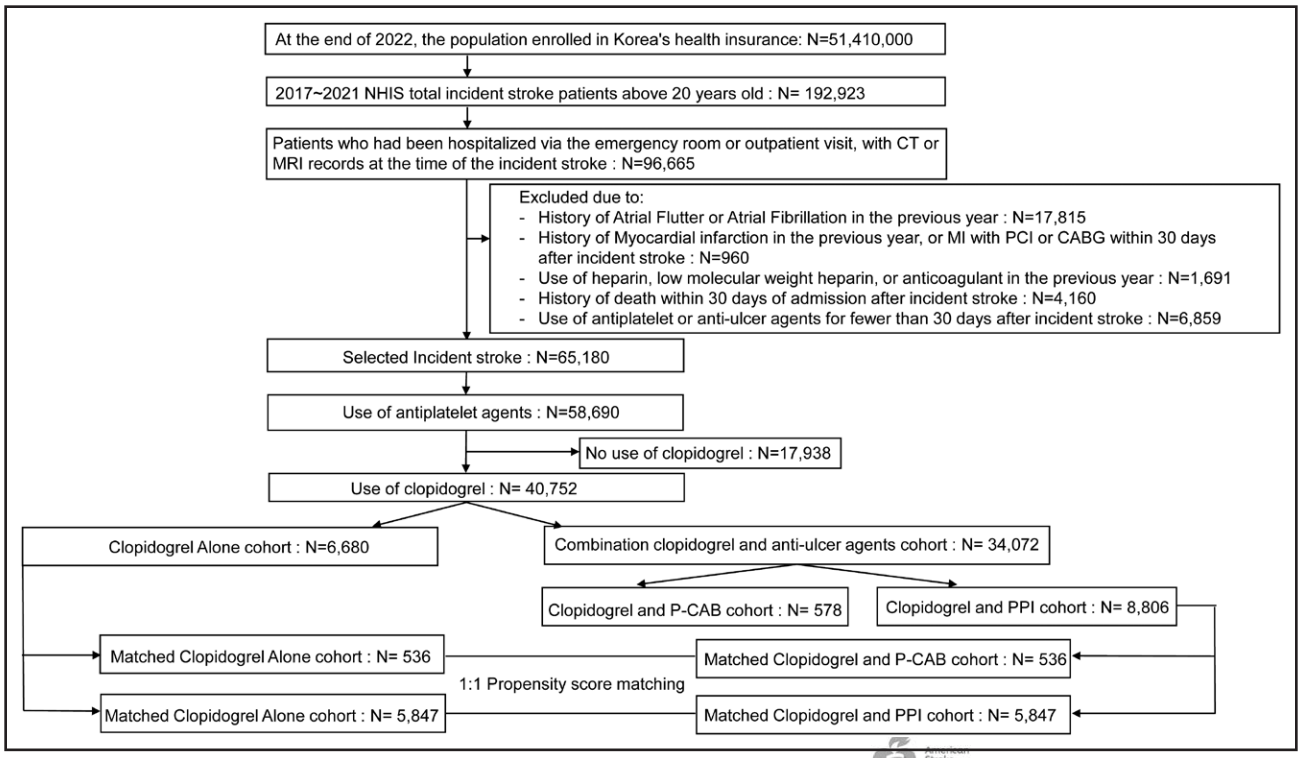


**Figure. Process for selecting the study population.**
CABG indicates coronary artery bypass grafting; CT, computed tomography; MI, myocardial infarction; NHIS, National Health Insurance Service; P-CAB, potassium-competitive acid blocker; PCI, percutaneous coronary intervention; and PPI, proton pump inhibitor.

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significantly higher in the clopidogrel+P-CAB cohort compared with the clopidogrel-alone cohort (HR, 2.64 [95% CI, 1.27–5.47]; Figure S2). No significant differences were observed for MI, death, or gastrointestinal bleeding between the 2 cohorts.

## Clopidogrel+PPI Cohort

The clopidogrel+PPI cohort experienced a significantly higher incidence of MACE than the clopidogrel-alone cohort in the PSM analysis (HR, 1.38 [95% CI, 1.17–1.63]; Table 4). Stroke recurrence was also significantly higher in the clopidogrel+PPI cohort than in the clopidogrel-alone cohort (HR, 1.41 [95% CI, 1.20–1.67]; Figure S3). Among the individual PPIs, only esomeprazole was associated with a significant increase in MACE (HR, 1.46 [95% CI, 1.10–1.92]; Table 5), accompanied by a higher risk of stroke recurrence (HR, 1.50 [95% CI, 1.13–2.00]). By contrast, no statistically significant differences in MACE were observed for the other PPIs between the clopidogrel+PPI cohort and the clopidogrel-alone cohort: (dex)lansoprazole (HR, 1.19 [95% CI, 0.94–1.50]; Table S5), omeprazole (HR, 0.44 [95% CI, 0.14–1.42]; Table S6), pantoprazole (HR, 0.91 [95% CI, 0.64–1.29]; Table S7), and ilaprazole or rabeprazole (HR, 1.22 [95% CI, 0.83–1.79]; Table S8). No statistically significant differences were observed in MI, death, or gastrointestinal bleeding between the clopidogrel-alone cohort and the clopidogrel+PPI cohort in the PSM analysis. In addition, a sex-stratified analysis revealed that females showed significantly elevated risks for both MACE (HR, 1.61 [95% CI, 1.24–2.10]) and stroke recurrence (HR, 1.58 [95% CI, 1.21–2.06]) in the clopidogrel+PPI cohort compared with the clopidogrel-alone cohort (Tables S9 and S10).

# DISCUSSION

This population-based, longitudinal cohort study showed that concomitant use of clopidogrel with P-CAB or PPI, especially esomeprazole, was associated with an increased risk of ischemic events. To the best of the authors' knowledge, this is the first study to show that P-CAB and esomeprazole may reduce the effectiveness of clopidogrel in patients with stroke.

In this study, the incidence of MACE was significantly higher in the clopidogrel+P-CAB cohort (26/536 patients, 4.85%) than in the clopidogrel-alone cohort (11/536 patients, 2.05%). Among MACE, the stroke recurrence rate was significantly higher in the clopidogrel+P-CAB cohort (26 patients, 4.85%) than in the clopidogrel-alone cohort (10 patients, 1.87%). This finding is consistent with a previous report indicating that, in a small study enrolling 31 healthy volunteers, one of the new P-CABs, vonoprazan, attenuated the antiplatelet effect of clopidogrel in the $P2Y_{12}$ assay. In that study,

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**Table 1. Demographic and Clinical Characteristics of Patients**

| | Clopidogrel alone (n=6680) | Clopidogrel+P-CAB (n=578) | Clopidogrel+PPI (n=8806) |
|---|---|---|---|
| Age,* y; mean (SD) | 67.74 (12.32) | 66.44 (12.98) | 68.14 (12.47) |
| Male sex, n (%) | 3984 (59.64) | 327 (56.57) | 5022 (57.03) |
| CCI, mean (SD) | 1.51 (0.98) | 1.4 (0.88) | 1.54 (0.96) |
| CCI category | | | |
| 0 | 68 (1.02) | 7 (1.21) | 70 (0.79) |
| 1 | 5024 (75.21) | 460 (79.58) | 6472 (73.5) |
| ≥2 | 1588 (23.77) | 111 (19.20) | 2264 (25.71) |
| Previous diagnosis, n (%) | | | |
| Hemorrhagic stroke | 63 (0.94) | 5 (0.87) | 88 (1.00) |
| Gastrointestinal bleeding | 10 (0.15) | 0 (0.00) | 25 (0.28) |
| Gastroesophageal reflux disease | 10 (0.15) | 43 (7.44) | 25 (0.28) |
| Peptic ulcer | 162 (2.43) | 7 (1.21) | 261 (2.96) |
| Gastritis and duodenitis | 478 (7.16) | 36 (6.23) | 720 (8.18) |
| Heart failure | 124 (1.86) | 14 (2.42) | 196 (2.23) |
| Angina | 261 (3.91) | 24 (4.15) | 495 (5.62) |
| Essential hypertension | 4235 (63.40) | 355 (61.42) | 5824 (66.14) |
| Dyslipidemia | 3297 (49.36) | 290 (50.17) | 4848 (55.05) |
| diabetes | 2651 (39.69) | 213 (36.85) | 3773 (42.85) |
| COPD | 114 (1.71) | 5 (0.87) | 154 (1.75) |
| Chronic renal failure | 223 (3.34) | 18 (3.11) | 387 (4.39) |
| Previous use, n (%) | | | |
| Aspirin | 500 (7.49) | 86 (14.88) | 1526 (17.33) |
| β-Blocker | 576 (8.62) | 57 (9.86) | 1166 (13.24) |
| ACE inhibitor | 64 (0.96) | 6 (1.04) | 142 (1.61) |
| ARB | 839 (12.56) | 103 (17.82) | 1744 (19.8) |
| Antihypertensive combination | 527 (7.89) | 74 (12.8) | 948 (10.77) |
| Diuretic antihypertensive | 36 (0.54) | 6 (1.04) | 71 (0.81) |
| Calcium-channel blocker | 862 (12.9) | 90 (15.57) | 1696 (19.26) |
| Other antihypertensive | 45 (0.67) | 5 (0.87) | 101 (1.15) |
| SSRI | 425 (6.36) | 51 (8.82) | 861 (9.78) |
| Hormonal agent | 37 (0.55) | 3 (0.52) | 61 (0.69) |

ACE indicates angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; CCI, Charlson Comorbidity Index; COPD, chronic obstructive pulmonary disease; P-CAB, potassium-competitive acid blocker; PPI, proton pump inhibitor; and SSRI, selective serotonin reuptake inhibitor.

*At the time of the first incident stroke occurred during clopidogrel therapy.

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among homogeneous extensive CYP2C19 metabolizers, the median inhibition of platelet aggregation for clopidogrel users was 48%, whereas that for clopidogrel plus vonoprazan users was 21%. In poor CYP2C19 metabolizers, the corresponding median inhibition values were 17% for clopidogrel users and 2% for clopidogrel plus vonoprazan users, respectively.[9]

From a pharmacokinetic perspective, vonoprazan is metabolized via multiple enzymatic pathways involving several CYP (cytochrome P450) isoforms (CYP3A4/5, CYP2B6, CYP2C19, CYP2C9, and CYP2D6) and non-CYP enzymes such as sulfotransferases and glucuronosyltransferases.[8,22] In a study of 7 healthy male volunteers, esomeprazole and vonoprazan significantly reduced the conversion of proguanil, a CYP2C19 substrate, to its active metabolite cycloguanil, as evidenced by a marked decrease in the AUC ratio (cycloguanil/proguanil).[23] The AUC ratio was reduced to 0.317-fold (95% CI, 0.256–0.379) in the esomeprazole group and 0.507-fold (95% CI, 0.409–0.605) in the vonoprazan group. Therefore, vonoprazan may competitively inhibit the conversion of clopidogrel into its active metabolite and reduce the antiplatelet effect of clopidogrel.[23]

This appears to be the first study to examine the increased risk of ischemic events in patients with stroke treated with a combination of clopidogrel and P-CABs. Because approval of P-CABs occurred near the end of the study period, the number of patients with follow-up

**Table 2. Balance of Covariates After 1:1 Propensity Score Matching**

| Patient characteristics | Clopidogrel alone (n=536) | Clopidogrel+P-CAB (n=536) | SMD | Clopidogrel alone (n=5847) | Clopidogrel+PPI (n=5847) | SMD |
|---|---|---|---|---|---|---|
| Sex, n | | | | | | |
| Male | 306 | 306 | 0.0000 | 3497 | 3505 | 0.0014 |
| Female | 230 | 230 | | 2350 | 2342 | |
| Age, y; n | | | 0.0074 | | | 0.0047 |
| 20 | 3 | 3 | | 10 | 10 | |
| 30 | 11 | 11 | | 71 | 70 | |
| 40 | 39 | 39 | | 369 | 369 | |
| 50 | 96 | 96 | | 995 | 991 | |
| 60 | 125 | 125 | | 1472 | 1470 | |
| 70 | 177 | 178 | | 1896 | 1899 | |
| 80 | 85 | 84 | | 1034 | 1038 | |
| CCI, n | | | 0.0056 | | | 0.0015 |
| 0 | 6 | 6 | | 57 | 54 | |
| 1 | 438 | 437 | | 4499 | 4500 | |
| ≥2 | 92 | 93 | | 1291 | 1293 | |
| Heart failure, n | 3 | 3 | 0.0000 | 58 | 59 | −0.0002 |
| GI bleeding, n | 0 | 0 | 0.0000 | 1 | 1 | 0.0000 |
| Hemorrhagic stroke, n | 1 | 1 | 0.0000 | 27 | 27 | 0.0000 |
| Hypertension, n | 185 | 186 | −0.0019 | 1684 | 1684 | 0.0000 |
| Dyslipidemia, n | 10 | 10 | 0.0000 | 91 | 93 | −0.0003 |
| Diabetes, n | 87 | 88 | −0.0019 | 1191 | 1192 | −0.0002 |
| COPD, n | 3 | 3 | 0.0000 | 61 | 62 | −0.0002 |
| GERD, n | 30 | 30 | 0.0000 | 293 | 300 | −0.0012 |
| Peptic ulcer, n | 3 | 3 | 0.0000 | 71 | 72 | −0.0002 |
| Gastritis and duodenitis, n | 25 | 24 | 0.0019 | 304 | 300 | 0.0007 |
| Angina, n | 19 | 18 | 0.0019 | 156 | 164 | −0.0014 |
| Chronic renal failure, n | 11 | 11 | 0.0000 | 151 | 150 | 0.0002 |
| Aspirin, n | 211 | 211 | 0.0000 | 1479 | 1478 | 0.0002 |

CCI indicates Charlson Comorbidity Index; COPD, chronic obstructive pulmonary disease; GERD, gastroesophageal reflux disease; GI, gastrointestinal; P-CAB, potassium-competitive acid blocker; PPI, proton pump inhibitor; and SMD, standardized mean difference.

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data was limited. Consequently, combined clopidogrel+P-CAB usage was analyzed rather than the usage of individual drugs. Future studies should reevaluate the clinical outcomes associated with each individual P-CAB.

A statistically significant difference in the incidence of MACE was observed between the clopidogrel-alone cohort (248/5847 patients, 4.24%) and the clopidogrel+PPI cohort (339/5847 patients, 5.80%). Stroke recurrence was also significantly more frequent in the clopidogrel+PPI cohort (330 patients, 5.64%) than in the clopidogrel-alone cohort (236 patients, 4.04%). This finding contrasts with a prior nested case–control study that found no association between clopidogrel alone and clopidogrel+PPI after stroke.[18] That study included 2765 patients aged ≥66 years,[18] whereas the present analysis examined 40 752 patients, all aged ≥20 years. Although this discrepancy may stem from factors such as genetic variation or differences in the individual PPIs used, the larger sample size and propensity score–matching methodology applied in this study may provide a more robust assessment of the influence of PPIs on clopidogrel effectiveness. Similarly, a previous claims database study showed that patients who underwent percutaneous coronary intervention with stent placement and received both clopidogrel and a PPI had a significantly higher risk of stroke within 12 months than did those given clopidogrel alone.[24] Because clopidogrel requires CYP2C19-mediated activation, coadministration of PPIs can inhibit CYP2C19 and reduce clopidogrel's antiplatelet activity.[25]

In this study, a subgroup analysis of individual PPIs was performed. Among PPIs, a statistically significant increase in MACE was observed only in the clopidogrel+esomeprazole cohort (121/2179 patients, 5.55%) compared with the clopidogrel-alone cohort (84/2179 patients, 3.85%). The risk of stroke recurrence likewise differed significantly between the clopidogrel+esomeprazole cohort (119 patients, 5.46%) and the clopidogrel-alone cohort (80 patients,

**Table 3. Crude and Propensity Score–Matched Analyses of MACE and GI Bleeding in Patients With Incident Stroke Receiving Clopidogrel With or Without P-CAB Coadministration**

| | Crude cohort | | | Propensity score–matched cohort | | |
|---|---|---|---|---|---|---|
| | Clopidogrel alone (n=6680) | Clopidogrel+P-CAB* (n=578) | | Clopidogrel alone (n=536) | Clopidogrel+P-CAB (n=536) | |
| | Events (%) | Events (%) | HR (95% CI) | Events (%) | Events (%) | HR (95% CI) |
| MACE | 282 (4.22%) | 26 (4.50%) | 1.10 (0.73–1.64) | 11 (2.05%) | 26 (4.85%) | 2.40 (1.18–4.85) |
| MI | 12 (0.18%) | 0 (0.00%) | | 1 (0.19%) | 0 (0.00%) | |
| Stroke recurrence | 268 (4.01%) | 26 (4.50%) | 1.16 (0.77–1.74) | 10 (1.87%) | 26 (4.85%) | 2.64 (1.27–5.47) |
| Death | 2 (0.03%) | 0 (0.00%) | | 0 (0.00%) | 0 (0.00%) | |
| GI bleeding | 23 (0.34%) | 1 (0.17%) | 0.50 (0.07–3.76) | 2 (0.37%) | 1 (0.19%) | 0.50 (0.05–5.50) |

GI indicates gastrointestinal; HR, hazard ratio; MACE, major adverse cardiovascular events; MI, myocardial infarction; and P-CAB, potassium-competitive acid blocker.
*P-CAB agents included vonoprazan, tegoprazan, fexuprazan, and revaprazan.

3.67%). This finding is consistent with FDA guidelines, indicating that esomeprazole may interact with clopidogrel in a manner similar to omeprazole.[3] However, the clopidogrel+omeprazole cohort comprised only 161 patients (1.83%) among all PPI cohorts. This small sample size reflects the limited prescription of omeprazole with clopidogrel in South Korea after the FDA recommendation and was insufficient to yield meaningful statistical results. In addition, rabeprazole has been consistently demonstrated to be safe in combination therapy across multiple studies.[11–14] In alignment with these findings, the current analysis showed no significant difference in MACE between the clopidogrel-alone cohort and the clopidogrel+ilaprazole/rabeprazole cohort (Table S11).

Significant differences among the clopidogrel+individual PPI cohorts may stem from variability in CYP2C19 inhibition across PPIs. A recent critical review showed that commonly used PPIs interact with CYP enzymes.[13] Consistent with the present findings, esomeprazole, a moderate CYP2C19 inhibitor, reduces clopidogrel activation, whereas rabeprazole, a weak CYP2C19 inhibitor, has little effect on clopidogrel activation.

In a meta-analysis of clopidogrel efficacy for acute ischemic stroke or TIA, CYP2C19 loss-of-function alleles were identified in 25% of White and 59% of Asian patients.[26] Supporting these data, a recent study of Korean patients demonstrated a high prevalence of the CYP2C19 loss-of-function allele (62.1%).[27] In a large randomized trial of 6412 patients with minor ischemic stroke or TIA carrying CYP2C19 loss-of-function alleles, ticagrelor was associated with a modestly lower risk of recurrent stroke at 90 days compared with clopidogrel; events occurred in 191 patients (6.0%) in the ticagrelor group and 243 patients (7.6%) in the clopidogrel group.[28] These findings further confirm that CYP2C19 loss-of-function alleles influence the effectiveness of clopidogrel. Also, the genetic polymorphism, such as the CYP2C19 genetic polymorphism or other genetic variants, may influence the interaction between clopidogrel and P-CABs or PPIs.[25] Therefore, future studies investigating the clinical impact of drug-drug interactions between clopidogrel and concomitant PPIs or P-CABs should incorporate a genetic/pharmacogenomic analysis of relevant genetic variants.

Our study found a notable sex-specific difference in vascular outcomes associated with PPI use among clopidogrel-treated patients, with the adverse association observed exclusively in female patients. Women seem to be more vulnerable to clopidogrel therapeutic failure due to PPI interaction. A study showed that the prevalence of the CYP2C19 poor metabolizer genotype is higher in the female population.[29] Because genetic

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**Table 4. Crude and Propensity Score–Matched Analyses of MACE and GI Bleeding in Patients With Incident Stroke Receiving Clopidogrel With or Without PPI Coadministration**

| | Crude cohort | | | Propensity score–matched cohort | | |
|---|---|---|---|---|---|---|
| | Clopidogrel alone (n=6680) | Clopidogrel+PPI* (n=8806) | | Clopidogrel alone (n=5847) | Clopidogrel+PPI (n=5847) | |
| | Events, % | Events, % | HR (95% CI) | Events, % | Events, % | HR (95% CI) |
| MACE | 282 (4.22%) | 443 (5.03%) | 1.15 (0.99–1.34) | 248 (4.24%) | 339 (5.80%) | 1.38 (1.17–1.63) |
| MI | 12 (0.18%) | 8 (0.09%) | 0.52 (0.21–1.27) | 10 (0.17%) | 5 (0.09%) | 0.50 (0.17–1.46) |
| Stroke recurrence | 268 (4.01%) | 431 (4.89%) | 1.18 (1.01–1.37) | 236 (4.04%) | 330 (5.64%) | 1.41 (1.20–1.67) |
| Death | 2 (0.03%) | 5 (0.06%) | 1.82 (0.35–9.52) | 2 (0.03%) | 4 (0.07%) | 2.00 (0.37–10.92) |
| GI bleeding | 23 (0.34%) | 32 (0.36%) | 0.97 (0.57–1.67) | 20 (0.34%) | 18 (0.30%) | 0.90 (0.48–1.70) |

GI indicates gastrointestinal; HR, hazard ratio; MACE, major adverse cardiovascular events; MI, myocardial infarction; and PPI, proton pump inhibitor.
*PPIs included esomeprazole, (dex)lansoprazole, omeprazole, pantoprazole, ilaprazole, and rabeprazole.

**Table 5. Crude and Propensity Score–Matched Analyses of MACE and GI Bleeding in Patients With Incident Stroke Receiving Clopidogrel With or Without Esomeprazole Coadministration**

| | Crude cohort | | | Propensity score–matched cohort | | |
|---|---|---|---|---|---|---|
| | Clopidogrel alone (n=6680) | Clopidogrel+ esomeprazole (n=2482) | | Clopidogrel alone (n=2179) | Clopidogrel+ esomeprazole (n=2179) | |
| | Events (%) | Events (%) | HR (95% CI) | Events (%) | Events (%) | HR (95% CI) |
| MACE | 282 (4.22) | 133 (5.36) | 1.24 (1.00–1.53) | 84 (3.85) | 121 (5.55) | 1.46 (1.10–1.92) |
| MI | 12 (0.18) | 2 (0.08) | 0.49 (0.11–2.22) | 4 (0.18) | 2 (0.09) | 0.50 (0.09–2.73) |
| Stroke recurrence | 268 (4.01) | 131 (5.28) | 1.28 (1.04–1.59) | 80 (3.67) | 119 (5.46) | 1.50 (1.13–2.00) |
| Death | 2 (0.03) | 1 (0.04) | 1.58 (0.14–18.35) | 0 (0.00) | 1 (0.05) | |
| GI bleeding | 23 (0.34) | 10 (0.40) | 1.12 (0.53–2.38) | 8 (0.37) | 9 (0.41) | 1.13 (0.43–2.92) |

GI indicates gastrointestinal; HR, hazard ratio; MACE, major adverse cardiovascular events; and MI, myocardial infarction.

information, such as CYP2C19 polymorphism, is not available in our study, we do not know whether the sex difference in vascular outcome is caused by genotype differences or not.

On the other hand, there was no difference in vascular outcome between men and women according to P-CAB use in patients treated with clopidogrel (Tables S12 and S13). However, this interpretation should be viewed cautiously due to the relatively small sample size of this group. Future research is warranted to investigate the influence of genetic information or sex differences on the interaction between clopidogrel and PPI/P-CAB.

## Limitations

Within the present cohort, MI incidence was relatively low in the clopidogrel-alone and clopidogrel+PPI cohorts (0.18% and 0.09%, respectively). The small sample size of the clopidogrel+P-CAB cohort also precluded meaningful comparison, underscoring the need for further research. Patients with a history of MI, atrial flutter, or atrial fibrillation within the previous year were excluded, resulting in a limited number with prior coronary artery disease. A linkage study of acute MI risk after stroke found that the incidence of acute MI in patients without preexisting cardiovascular disease during the first year after stroke was 0.8%.[30] Given that the follow-up period in this study was 6 months, the observed low MI incidence appears consistent with that report. Finally, it must be acknowledged that P-CABs are relatively newly developed agents, and while they have been approved and are available in specific regions such as the United States, South Korea, and Japan, they are not available worldwide. Considering that antiulcer medications are common in clinical practice, the results of this study may hold significant clinical implications not only for stroke specialists but also for clinicians in a wide range of specialties, such as internal medicine and family medicine. Therefore, further studies will be required to confirm whether our findings can be consistently reproduced in other geographic regions or among diverse ethnic groups.

## Conclusions

Concomitant use of a PPI or P-CAB was associated with an increased risk of ischemic events among clopidogrel-treated stroke survivors. When a PPI is required, esomeprazole should not be considered the first-line agent because it was associated with the greatest increase in MACE and stroke recurrence, potentially owing to stronger inhibition of clopidogrel activation. Likewise, the possible impact of P-CABs on MACE and stroke recurrence must be weighed when initiating clopidogrel plus P-CAB therapy. Clinicians should exercise caution when prescribing a PPI or P-CAB to individuals receiving clopidogrel.

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## ARTICLE INFORMATION

Received September 2, 2025; final revision received January 25, 2026; accepted February 4, 2026.

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### Acknowledgments

The authors sincerely thank the National Health Insurance Service for providing access to the Korean National Health Insurance Database, which was essential for this study.

### Sources of Funding

This work was supported by the Republic of Korea Government through multiple grants: the Ministry of Science and Information and Communication Technology (National Research Foundation of Korea, NRF-2021R1F1A1062044), the Basic Science Research Program of the National Research Foundation of Korea funded by the Ministry of Education (2021R1A6A1A03044296), and the Ministry of Food and Drug Safety (22183MFDS366). The funders had no role in study design, data collection, analysis, interpretation, or manuscript preparation.

### Disclosures

None.

### Supplemental Material

Tables S1–S13
Figures S1–S3
STROBE Checklist

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